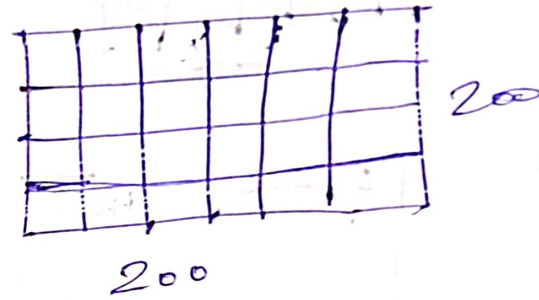
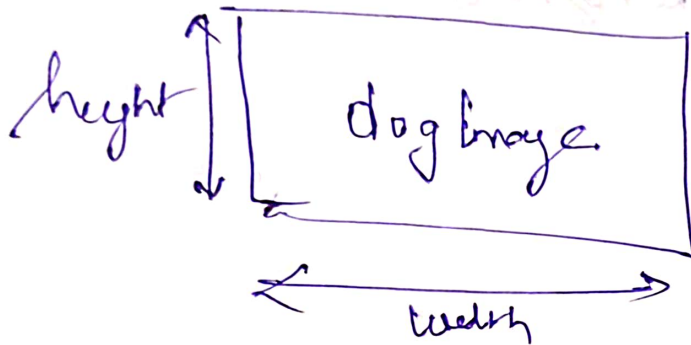


Q1

Image processing

Image dimension :-



Eg 200×200
width $\times$ height

200 boxes in horizontal axis
 $200 \rightarrow 4$ units
4

$$\begin{aligned} \text{Total pixel} &= 200 \times 200 \\ &= 40,000 \end{aligned}$$

Image resolution :-

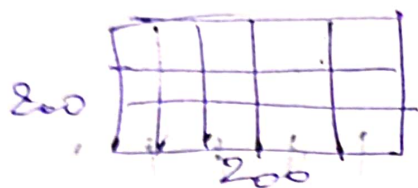
480p $\Rightarrow 1280 \times 480$
720p $\Rightarrow 1280 \times 720$
1080p $\Rightarrow 1920 \times 1080$
4k $\Rightarrow 3840 \times 2160$
8k $\Rightarrow 7680 \times 4320$

Types of Image

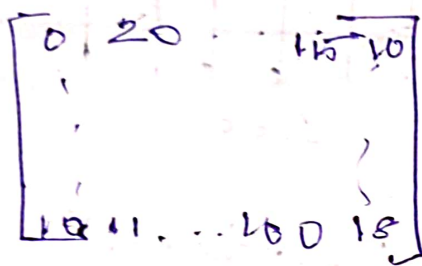
Grayscale $\rightarrow$ Black & White $\Rightarrow$ 1 channel

RGB $\rightarrow$ 3 channel $\rightarrow$ Red
Green
Blue

How to convert this grayscale to
Numerical value?



Total pixel $\Rightarrow 400000$



200x200

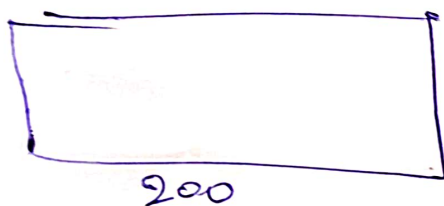
Color range:

[0 255]

black

white

RGB Images-

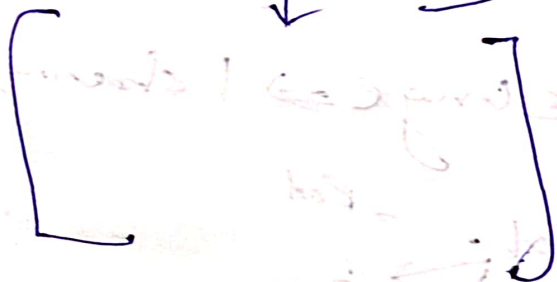


$\Rightarrow 400000$

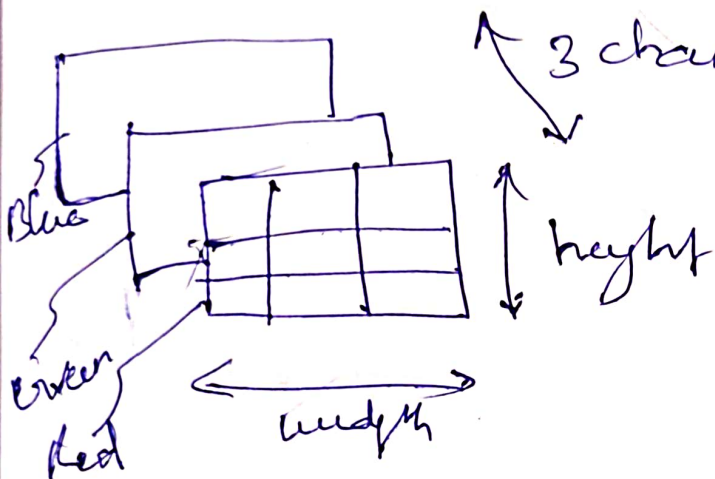
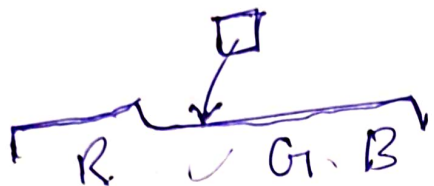
Red color intensity

Green

Blue



for particular ^{near} pixel, $\rightarrow$ range $[0, 255]$



$$\begin{cases} R = 255 \\ G = 255 \\ B = 255 \end{cases} \Rightarrow \text{white}$$